

Supplemental: On the Value of Non-i.i.d. Tabular IoT Data for Federated Learning

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1 Execution Time Analysis

The adaptation of Shapley Values (SV) to federated learning (FL) [Wang et al. 2020] is a method that calculates how much a participant contributes to a training session. However, it also has a computational complexity of 2^n , where n is the number of participants. This makes the SV method impractical for scenarios with low computational resources or many clients. In this Appendix, we analyze how much overhead in time the SV computation adds to the training process.

First, we plot the time overhead added by the SV computation in both centralized and decentralized settings. The results show the mean overhead per round for each configuration. This is represented by Figure 1. As can be seen, the computation adds tremendous overhead, increasing up to ten times the amount of time needed for a round of training. We also do not notice much difference in the aggregation process, at least per round. The main cause of overhead seems to be the dataset itself. This is reinforced by Figure 2, which presents a linear increase in execution time based on size. The high variability in training time is due to the early stop functionality included in the training. Then, in Figure 3, we can examine the relative time each process occupies in a round. The process that takes most of the time, on average 60-80%, is the centralized SV computation. The decentralized version took only 20-25% of the time. We think this is caused by parallelizing the training and having smaller datasets on each client. However, it still takes considerable time. Finally, training takes 10-15% of the time. The difference between training and SV seems smaller the faster the training is performed, which seems to align with having a higher overhead in loading different models than in the inference process itself.

We go into more detail about the full execution times for each training. Figure 4 presents the total execution time for training, based on aggregation processes and datasets. The first obvious conclusion is that the training scales linearly with the number of rounds. Three groups can be seen in Figure 4, based on the shape of the points. HAR, with circles, occupies the bottom row. Then, wids_dataset occupies a middle place with diamonds. Finally, squares represent the edge-iot-coreset and take the most time. And each of the

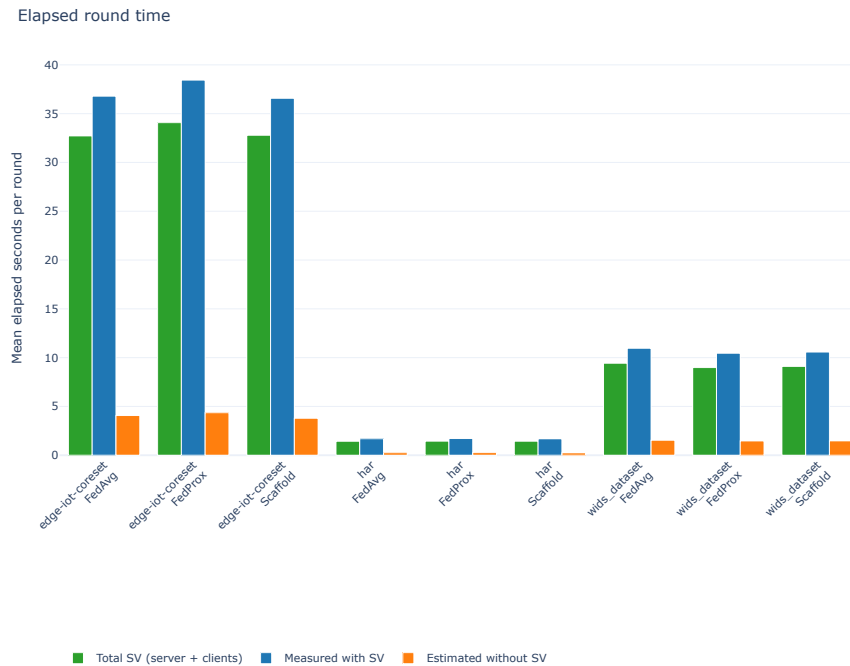


Figure 1: Bar plot for the mean computation time per round. The groups are the aggregation process and the dataset. The mean value is computed for all partitions and random seeds.

groups follows such linear scaling. The other three factors determining time are: the dataset size, which was already explored; how evenly such data was partitioned, as this affects how much time the last client computing local SV will take; and the number of rounds. For the latter, we have not found any aggregation process that speeds up convergence enough to trigger the early termination process.

References

[Wang et al. 2020] Wang, T., Rausch, J., Zhang, C., Jia, R., Song, D.: "A Principled Approach to Data Valuation for Federated Learning"; in Yang, Q., Fan, L., Yu, H. (eds.): Federated Learning: Privacy and Incentive, Lect. Notes Comp. Sci. 12500, Springer International Publishing, Cham (2020), 153–167.

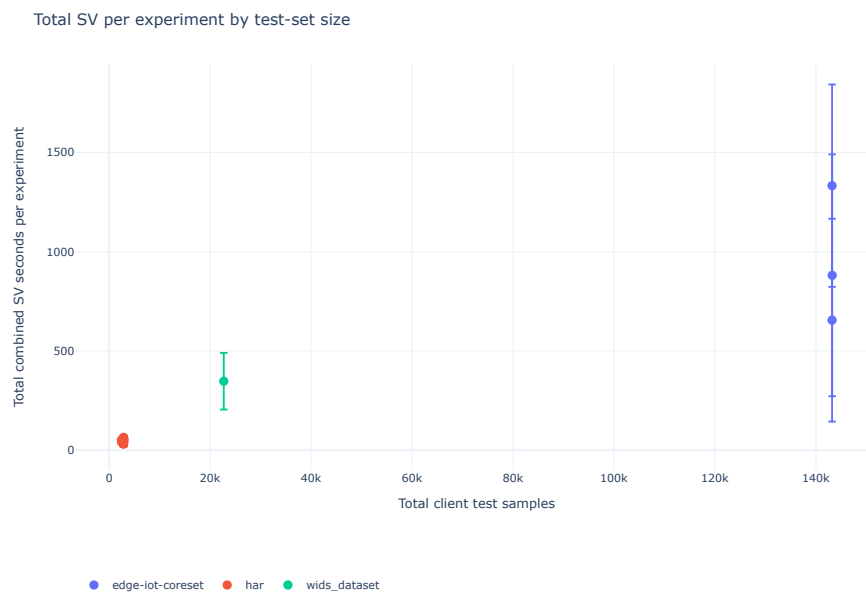


Figure 2: Analysis of scalability based on total time for dataset size.

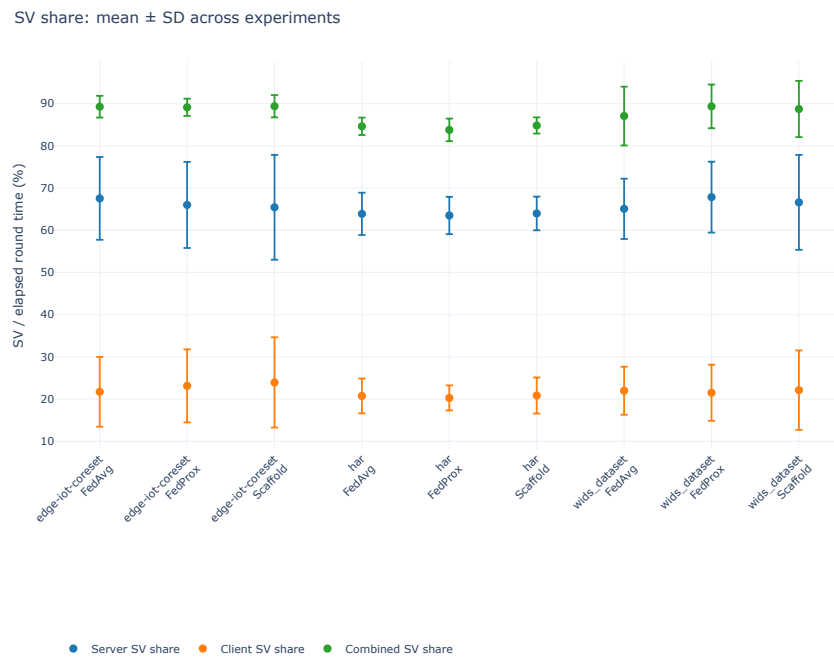


Figure 3: Analysis of the percentage of time spent in each of the SV computations, plus the total percentage of the time occupied by the SV computation.



Figure 4: Analysis of the total training time based on the number of rounds.